Hardware-Aware Speculative KV-Cache Paging and Epistemic Attention Eviction for High-Throughput Multi-Turn Agentic LLM Serving

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***Abstract*—The rapid proliferation of autonomous multi-turn agentic workflows has exacerbated the memory capacity bottleneck in Large Language Model (LLM) inference accelerators. In modern agentic sequences spanning 32k to 128k+ tokens ($L = 64, H_{\text{kv}} = 8, d_k = 128$, `float16`), Key-Value (KV) cache memory footprints expand to 17.18 GB (16.00 GiB) per active 64k-token agent stream (8.59 GB / 8.00 GiB at 32k tokens). Even under multi-chip tensor-parallel weight and KV-cache sharding or 32k–64k context distributions, this memory demand precipitates an acute capacity cliff on accelerators such as the Google Cloud TPU v6e Trillium (32 GB HBM per chip), exhausting non-weight HBM capacity after only 3 to 8 concurrent sequences. Existing virtualized inference engines (e.g., PagedAttention / vLLM and SGLang / RadixAttention) mitigate Out-Of-Memory exceptions via reactive demand paging to Host DRAM over PCIe/CXL. However, servicing a synchronous full-layer demand page fault (64×64 KiB $= 4$ MiB across $L = 64$ layers, or $1.87\,\mu$s of raw wire serialization per single-layer 64 KiB block at 35 GB/s effective PCIe Gen5 bandwidth plus descriptor ring wakeup and dirty writeback) incurs severe pipeline stalls (0.12 ms to 0.50 ms per faulted step against 0.16 ms ideal compute), causing matrix execution units to idle for 40% to 70% of execution time. Furthermore, classical page eviction policies (LRU/LFU) are attention-blind: they inadvertently evict initial prompt attention sinks and active tool schemas while preserving ephemeral chain-of-thought scratchpad tokens.**

To overcome these constraints, we present Speculative Epistemic KV-Cache Paging (SE-Paging) governed by the Epistemic Least-Likely-Attended (ELLA) eviction protocol. SE-Paging couples the agent's semantic execution graph and turn-smoothed epistemic branch entropy $\mathcal{H}_i(t)$ to physical memory management: (1) an Epistemic Working Set Estimator (EWSE) projects future attention mass $\mathcal{A}(P_k, t)$, permanently pinning attention sinks and retaining active/idle tool schemas and DAG-boosted history while decaying idle tool schemas below DAG-predicted history within ≈ 8 steps (≈ 1 turn) and aggressively evicting resolved scratchpad pages; (2) an Asynchronous DMA Speculative Prefetcher (ADSP) with sealed-page write-through host mirroring overlaps Host-to-Device page transfers with active matrix computation across a $\tau_{\text{lookahead}} = 2$-step horizon; and (3) ELLA establishes bounded competitive ratios relative to offline Belady MIN under both non-cascading trace-regular greedy eviction and randomized Predictive Marker phases. We evaluate SE-Paging in a downscaled $1 : 32$ trace-driven CPU-based physical tensor-buffer memory-hierarchy emulator (modeling a 2-layer 128 KiB slice per $B = 16$-token page, $N_{\text{dev}} = 96$ HBM pages [12.00 MiB], $N_{\text{host}} = 2{,}500$ DRAM pages [312.50 MiB], and 976 allocated physical pages [16 shared prefix + 48 private pages/stream, or 15,616 total tokens] across $K = 20$ streams) parameterized to scaled TPU v6e Trillium and PCIe Gen5 timing, complemented by a 6-turn live trace replay on Google Cloud Vertex AI (`gemini-2.5-flash`) and multi-turn SWE-bench and Gorilla API trace evaluations. Across 5,940 logical page-lookup cycles (11,880 two-stage physical page transactions per protocol) spanning $K \in \{4, 8, 12, 16, 20\}$ concurrent agent streams under $10.17\times$ physical ($13.33\times$ unshared logical) working-set overflow, SE-Paging elevates cache hit rate at $K = 20$ from 0.20% (LRU) and 1.21% (LFU) to 28.23% ($28.31 \pm 0.38\%$ 5-seed mean), retains 98.64% empirical Query-Key (QK) RoPE dot-product attention mass, eliminates 46.8% of memory pipeline stall overhead, and increases simulated device throughput by 49.3% (+53.4% in host wall-clock page-lookup throughput) over LRU.

***Index Terms*—KV-Cache Memory Management, Hardware Accelerators, TPU v6e Trillium, Speculative DMA Prefetching, Multi-Agent LLM Serving, PagedAttention, Virtual Memory.**

I. Introduction

Enterprise deployment of autonomous agent swarms built on foundation models (e.g., Gemini 2.5 Pro/Flash, Gemma 2, and Claude) represents a major workload shift in cloud computing. In frameworks such as the Google Agent Development Kit (ADK) and Vertex AI Agent Builder, agents engage in prolonged, multi-turn reasoning workflows involving hierarchical sub-goal decomposition, external tool invocations via OpenAPI schemas [18], and structured artifact generation [1], [2].

A. The KV-Cache HBM Capacity Cliff

Unlike traditional stateless NLP classification or single-turn dialogue, agentic sequences accumulate extensive conversational histories, tool schemas, and chain-of-thought scratchpads, routinely expanding context lengths from 32k to 128k+ tokens [3], [4].

During auto-regressive transformer decoding, Key and Value activation vectors for every past token must remain memory-resident to avoid redundant quadratic recomputation. The physical memory required scales linearly with context length S, transformer layers L, KV heads H_{kv}, and head dimension d_k:

$$\text{Memory}_{\text{KV}} = 2 \times L \times H_{\text{kv}} \times d_k \times S \times \text{bytes_per_elem} \quad (1)$$

For a representative enterprise model ($L = 64$, $H_{\text{kv}} = 8$, $d_k = 128$, `float16` with bytes_per_elem $= 2$), a sequence of $S = 64\text{k} = 65{,}536$ tokens requires 17,179,869,184 bytes $\approx$

17.18 GB decimal (**16.00** GiB binary) of dedicated High-Bandwidth Memory (HBM) (or $8{,}589{,}934{,}592$ bytes $\approx$ **8.59** GB decimal / **8.00** GiB binary for $S = 32\text{k} = 32{,}768$ tokens).

On modern cloud accelerators such as the Google Cloud TPU v6e Trillium (configured with 32 GB HBM in 256-chip 2D torus pods, following Google's custom TPU architectures [5]) or GPU nodes [6], this memory demand triggers an immediate *capacity cliff*: even under multi-chip tensor-parallel model-weight and KV-cache sharding (where 64-layer weights are partitioned across chips, leaving 12–24 GiB of non-weight HBM per chip) or 32k–64k context distributions with shared prefixes, a chip exhausts its remaining HBM pool after serving only 3 to 8 concurrent agent sequences, capping cluster concurrency.

B. The Demand Paging Pipeline Stall Dilemma

To prevent fatal Out-Of-Memory (OOM) aborts, modern serving runtimes (e.g., vLLM / PagedAttention [3] and SGLang [1]) partition the KV-cache into discrete non-contiguous physical pages ($B = 16$ tokens per block) and implement reactive *demand swapping* to Host DRAM over PCIe Gen5 or CXL, while multi-turn offloading systems such as CachedAttention [2] perform coarse-grained inter-turn session pre-loading across conversation boundaries.

However, intra-turn demand swapping is fundamentally reactive: when the accelerator's matrix execution units encounter an unmapped page during token generation, the decoding pipeline stalls. Over a PCIe Gen5 $\times 16$ interconnect (64 GB/s theoretical unidirectional bandwidth, $\approx$ 35 GB/s effective under multi-stream contention), transferring a single-layer 64 KiB page block ($B = 16, H_{\text{kv}} = 8, d_k = 128$) requires $\approx 1.87\,\mu$s of raw wire serialization ($3.74\,\mu$s per fused 2-layer 128 KiB slice, or $119.7\,\mu$s across all $L = 64$ transformer layers totaling 4 MiB per 16-token block); combined with PCIe DMA descriptor ring wakeup and synchronous dirty-page writeback, a full-model page fault stalls token generation for 0.12 **ms to** 0.50 **ms** (0.288–0.384 ms in our calibrated parameterization vs. 0.160 ms ideal 64-layer compute) [6], [7]. Under high sequence concurrency, these synchronous demand stalls accumulate, causing matrix execution units to spend **40% to 70% of wall-clock time stalled waiting on memory**.

C. The Attention-Blind Eviction Failure

Furthermore, classical operating system page eviction policies (Least Recently Used [LRU], Least Frequently Used [LFU], FIFO) and existing token-pruning or single-step rehearsal heuristics are mismatched with multi-turn agentic transformer physics:

1) **Destruction of Attention Sinks and Tool Schemas:** Transformers concentrate massive attention mass on the initial 1 to 4 sink blocks [4] and active OpenAPI tool definitions [18]. Under LRU, as an agent generates thousands of intermediate reasoning tokens, initial prompt tokens and tool schemas are aged out to Host RAM. While static window methods like StreamingLLM [4] drop middle tokens permanently, Heavy-Hitter Oracle (H2O) [8] tracks cumulative attention scores, Quest [9] estimates page criticality from query-key bounding boxes within the current layer without multi-step DMA prefetching, and InfiniGen [7] performs single-step layer-adjacent rehearsal prefetching (using skewed query-key projections in layer $l-1$ to prefetch layer l), none of these approaches satisfy multi-step PCIe lookahead horizons ($\tau_{\text{lookahead}} = 2$ steps ahead) during abrupt multi-turn agentic tool-call DAG transitions where an entire dormant tool schema or prior history segment suddenly regains high query-key attention alignment. Similarly, unlike purely static agentic workflow prefetchers (KVFlow [10]) or prefix-tree LRU schedulers (SGLang / RadixAttention [1]) that ignore live generation uncertainty, SE-Paging couples workflow DAG lookahead with turn-smoothed epistemic Shannon entropy $\mathcal{H}_i(t)$.
2) **Scratchpad Memory Pollution:** Intermediate chain-of-thought scratchpad tokens are predominantly transient: once an agent concludes a reasoning sub-step, its scratchpad tokens carry rapidly decaying query-key weight unless backtracking occurs. Classical LRU retains newly written scratchpad pages in fast HBM, displacing reusable history and tool schemas.

D. Contributions

In this paper, we propose **Hardware-Aware Speculative KV-Cache Paging (SE-Paging)** governed by the **Epistemic Least-Likely-Attended (ELLA)** eviction protocol. Our key contributions are:

1) **Epistemic Working Set Estimator (EWSE):** A formal priority formulation that projects future page attention mass $\mathcal{A}(P_k, t)$ from agent semantic segment tags, stream-indexed tool-call and history DAG lookahead $\hat{d}_{\text{DAG},i}(P_k, t)^{-1}$, and conditionally re-normalized action-delimiter epistemic Shannon entropy $\mathcal{H}_i(t)$.
2) **Asynchronous DMA Speculative Prefetcher (ADSP):** A non-blocking prefetcher restricted to lookahead-predicted non-resident candidates with sealed-page write-through host mirroring that dispatches Host-to-Device transfers at step $n - \tau_{\text{lookahead}}$, overlapping PCIe/CXL transfer latency with matrix compute while eliminating synchronous dirty-page writebacks.
3) **Theoretical Performance Guarantees:**
   - **Theorem 1 (Queued Zero-Stall Pipeline Overlap):** Establishes the step-indexed multi-page lookahead condition $\tau_{\text{lookahead}} \geq \left\lceil \frac{M_{\text{pre}}(n) T_{\text{DMA}}(P_k)}{(1-\rho_{\text{bus}}) T_{\text{compute}}} \right\rceil$ under an unbacklogged or steady-state stable DMA queue under which device stall cycles are identically zero.
   - **Theorem 2 (Attention Mass Preservation):** Proves that ELLA scratchpad eviction beyond the pinned active window ($\delta = 64$ tokens) of the executing stream analytically bounds attention mass loss by $\epsilon_{\text{attn}} \leq C\delta^{1-\gamma}\left(\frac{1}{\delta} + \frac{1}{\gamma-1}\right) \leq \frac{C_{\text{scratch}}}{(\gamma-1)\delta^{\gamma-1}}$ (≤ 0.0261

at $\delta = 64$ and ≤ 0.0105 at $\delta = 256$ from the intermediate integral bound), complemented by non-circular empirical Query-Key (QK) RoPE softmax attention evaluation showing 98.64% retained attention mass (96.80% top-k QK page recall).

- **Theorem 3 (Bounded Competitive Ratio):** Uses the competitive caching with predictions framework [11] on acyclic agentic execution graphs to establish expected competitive ratios of $\mathbb{E}[\mathcal{C}_{\text{ELLA}}^{\text{greedy}}] \leq 1+\mathcal{O}(\eta_{\text{tot}}/\text{OPT}+\mathcal{H}_{\max})$ under non-cascading trace-regular greedy eviction and $\mathbb{E}[\mathcal{C}_{\text{ELLA}}^{\text{marker}}] \leq 2 + \mathcal{O}(\min(\sqrt{\mathbb{E}[\eta_{\ell_1}]/\text{OPT}}, \log N_{\text{dev}}))$ when paired with randomized Predictive Marker phases relative to offline Belady MIN [12].

4) **Downscaled Physical Tensor-Buffer Emulator, Multi-Turn Benchmark Generalization & Live Vertex AI Trace Replay:** We evaluate SE-Paging in a downscaled $1:32$ CPU-based physical tensor-buffer emulator (modeling a 2-layer 128 KiB slice per page, $N_{\text{dev}} = 96$ HBM pages [12.00 MiB], $N_{\text{host}} = 2{,}500$ DRAM pages [312.50 MiB], and 976 allocated physical pages [16 shared prefix + 48 private pages/stream, or 15,616 total tokens] across $K = 20$ streams) across 5,940 logical page-lookup cycles (11,880 two-stage physical page transactions per protocol), comparing against 10 baseline and ablation configurations (`DEMAND_LRU`, `DEMAND_LFU`, `SGLang/RadixAttention` [1], `StreamingLLM` [4], `H2O` [8], `Quest` [9], `CachedAttention` [2], `InfiniGen` [7], `DEMAND_ELLA`, and `STATIC_DAG_ELLA` [10]), plus a $\pm 50\%$ hyperparameter sensitivity study, multi-turn SWE-bench [17] and Gorilla [18] workload traces, and a 6-turn live Google Cloud Vertex AI (`gemini-2.5-flash`) trace replay.

II. Related Work

- **Virtualized KV-Cache and Offloading Systems:** PagedAttention / vLLM [3] introduced OS-inspired block-level KV-cache virtualization ($B = 16$ tokens) with reactive demand swapping, while SGLang [1] introduced RadixAttention for shared prefix-tree reuse with LRU leaf eviction. FlexGen [6] formulated static linear-programming offload schedules across GPU, CPU DRAM, and disk for offline batch throughput, and CachedAttention [2] overlapped inter-turn session restoration across multi-turn conversation boundaries. However, none of these systems prevent intra-turn page-fault stalls when an active agent dynamically switches between tool schemas and history blocks across sub-steps.
- **Token Pruning and Intra-Layer Sparse Attention:** StreamingLLM [4] retains initial attention sinks plus a static sliding window while permanently discarding middle tokens. H2O [8] evicts tokens with low cumulative attention scores, Quest [9] selects top-k pages per layer using query-key bounding boxes without asynchronous Host-to-Device DMA lookahead, and InfiniGen [7] prefetches KV entries from layer $l-1$ to layer l via skewed projections. While effective for single-step layer overlap, single-layer lookahead ($\tau = 1$ layer) is insufficient to hide multi-page PCIe Gen5 transfer latency across multi-turn agentic state transitions.
- **Agent-Aware KV-Cache Management:** Recent agentic serving frameworks such as KVFlow [10] and SGLang [1] exploit static agentic workflow graphs or radix trees to pre-warm or pin agent prefixes during structured tool-calling workloads [17], [18]. However, static DAG distances are deterministic: when an agent encounters high epistemic branching uncertainty during reasoning or tool error recovery, static schedules either evict active scratchpad context prematurely or fail to adjust history retention dynamically. SE-Paging bridges this gap by combining fine-grained categorical page tagging ($\mathcal{P}_{\text{anchor}}, \mathcal{P}_{\text{tool}}, \mathcal{P}_{\text{active}}, \mathcal{P}_{\text{hist}}, \mathcal{P}_{\text{scratch}}$) and stream-indexed forward DAG distance $\hat{d}_{\text{DAG},i}(P_k, t)^{-1}$ with live turn-smoothed epistemic branch entropy $\mathcal{H}_i(t)$ and sealed-page write-through DMA mirroring.

III. System Architecture and Memory Hierarchy

A. Heterogeneous Memory Hierarchy Model

We model an AI inference accelerator node (e.g., Google Cloud TPU v6e Trillium programmed via JAX/Pallas kernels [13]) comprising two memory tiers:

- **Tier 0 (Device HBM):** High-bandwidth, capacity-constrained memory directly attached to Matrix Execution Units (MXUs) ($\text{BW}_{\text{HBM}} \geq 1.6$ TB/s, capacity $C_{\text{HBM}} \in \{32, 64\}$ GB).
- **Tier 1 (Host DRAM / CXL):** High-capacity system memory connected via PCIe Gen5 / CXL ($\text{BW}_{\text{bus}} \approx$ 32–64 GB/s, capacity $C_{\text{Host}} \geq 512$ GB).

KV-cache is managed in fixed-size physical page blocks P_k containing $B = 16$ tokens. For a single transformer layer ($H_{\text{kv}} = 8, d_k = 128$, `float16`), each page block occupies:

$$\begin{aligned}\text{Size}_{\text{1-layer}}(P_k) &= B \times 2 \times H_{\text{kv}} \times d_k \times 2 \text{ bytes} \\ &= 65{,}536 \text{ bytes } (64 \text{ KiB})\end{aligned} \tag{2}$$

In our multi-layer block allocator (Section 6.1), pairs of adjacent transformer layers are co-allocated into fused 128 KiB (131,072 byte) physical transfer units to amortize PCIe DMA descriptor ring overhead while permitting layer-pipelined DMA interleaving and sub-stack preemption across concurrent streams without blocking on a monolithic 4 MiB transfer. In a full 64-layer deployment, the 32 fused 2-layer slices of logical block P_k share a single logical page-table descriptor and priority score $\mathcal{A}(P_k, t)$ equipped with a 32-bit layer-residency bitmap (`slice_mask`); if sub-stack preemption interrupts a 64-layer transfer after slice $s < 32$, layers $1 \ldots 2s$ execute while the remaining slices resume at the head of the DMA queue prior to layer $2s + 1$. In our scaled $1:32$ slice emulator (modeling a representative 2-layer 128 KiB slice out of $L = 64$ layers), per-slice DMA wire transfer latency $T_{\text{DMA}}(P_k) = 3.74\,\mu$s, per-slice ideal matrix compute

time $T_{\text{compute}}^{\text{ideal}} = 5.00\,\mu$s, and scaled descriptor wakeup / writeback penalties ($T_{\text{wakeup}} = 4.44\,\mu$s, $T_{\text{wb}} = 3.00\,\mu$s) scale uniformly by $1:32$ relative to a full 64-layer step ($T_{\text{DMA}}^{\text{64L}} \approx 119.7\,\mu$s, $T_{\text{compute}}^{\text{64L}} = 160.0\,\mu$s, clean fault $288.0\,\mu$s, dirty fault $384.0\,\mu$s), preserving exact dimensionless overlap ratios, page-replacement hit rates, and working-set overflow ratios.

B. Categorical Page Classification and Boundary Resolution

As tokens are ingested or generated by the orchestrator (e.g., Google ADK or SGLang [1]), lightweight structural delimiters (<|system|>, <|tool_schema|>, <|history|>, <|thought|>) tag each token span in $\mathcal{O}(1)$ time with a persistent underlying structural class ($\mathcal{P}_{\text{anchor}}, \mathcal{P}_{\text{tool}}, \mathcal{P}_{\text{hist}}, \mathcal{P}_{\text{scratch}}$), while dynamically maintaining an active sliding window ($\mathcal{P}_{\text{active}}$):

1) **System Anchor ($\mathcal{P}_{\text{anchor}}$):** System prompts and invariant instructions acting as persistent attention sinks [4].
2) **Tool Schema ($\mathcal{P}_{\text{tool}}$):** OpenAPI function declarations and JSON schema contracts.
3) **Active Generation Window ($\mathcal{P}_{\text{active}}$):** A temporary sliding-window boost applied to the 4 most recent $B = 16$-token blocks ($W_{\text{local}} = 64$ tokens at block-aligned boundaries, or $3B + 1 = 49$ to 64 tokens when the unsealed tail block is partially populated) of the currently executing stream $S_{i^*(t)}$.
4) **Conversation History ($\mathcal{P}_{\text{hist}}$):** Verified multi-turn user queries and committed tool outputs.
5) **Intermediate Scratchpad ($\mathcal{P}_{\text{scratch}}$):** Ephemeral chain-of-thought reasoning tokens from completed turns.

Dominant-Class Boundary and Non-Demoting Sliding-Window Rule: Structural delimiter transitions seal the current block by default so ephemeral scratchpad tokens do not pollute higher-tier pages; if contiguous sub-block packing is enabled, any $B = 16$-token block straddling two delimiter spans dynamically promotes its persistent class $c(P_k) \in \{\mathcal{P}_{\text{anchor}}, \mathcal{P}_{\text{tool}}, \mathcal{P}_{\text{hist}}, \mathcal{P}_{\text{scratch}}\}$ under the dominance hierarchy $\mathcal{P}_{\text{anchor}} \succ \mathcal{P}_{\text{tool}} \succ \mathcal{P}_{\text{hist}} \succ \mathcal{P}_{\text{scratch}}$ prior to sealing. For the currently executing stream $S_{i^*(t)}$ at step t, any page P_k within its 4 most recent blocks ($W_{\text{local}} = 64$ tokens, indexed by block age $\Delta t_{\text{win}} \in \{0, 1, 2, 3\}$) is evaluated as $\tilde{\mathcal{A}}_i(P_k, t) = \max(\mathcal{A}_{\text{class},i}(P_k, t), \mathcal{A}_{\text{active},i}(P_k, t))$, ensuring that an active tool schema ($\mathbb{I}_{\text{tool},i} = 1$, score 5×10^8) or anchor (10^9) inside W_{local} is never demoted to 10^7, while history and scratchpad pages inside the executing stream's W_{local} are elevated to $\mathcal{A}_{\text{active},i}(P_k, t) = 10^7 \exp(-\kappa \Delta t_{\text{win}})$ during its attention computation and revert to $\mathcal{A}_{\text{class},i}(P_k, t)$ when the stream yields or advances past 4 blocks (preventing $K \times 4 = 80$ dormant stream windows from monopolizing the 96-page HBM arena).

IV. The Speculative Epistemic Paging Protocol

A. Epistemic Attention Scoring Formulation

Definition 1 (Projected Attention Score)**.** *Let $t \in \mathbb{N}$ denote the discrete execution / page-lookup step index (where each macroscopic agentic reasoning turn spans 8–12 execution steps). For a KV-cache page P_k evaluated for agent stream S_i at step t, let $\Delta t_i(P_k, t) = t - t_{\text{last_access},i}(P_k) \geq 0$ denote the elapsed execution step count since the semantic segment containing P_k was last generated, selected by the router, or accessed by stream S_i, and let $\hat{d}_{DAG,i}(P_k, t) \in \mathbb{N} \cup \{\infty\}$ denote the shortest-path forward step distance in stream S_i's workflow DAG from its current node to the next node referencing P_k's segment. Its per-stream class score $\mathcal{A}_{class,i}(P_k, t)$ under ELLA is defined as:*

$$\mathcal{A}_{class,i}(P_k, t) \triangleq \begin{cases} 10^9, & c(P_k) = \mathcal{P}_{anchor} \\ 5 \times 10^8\, \mathbb{I}_{tool,i}(P_k, t) + (1 - \mathbb{I}_{tool,i}(P_k, t))\left(W_{tool} e^{-\lambda_{tool} \Delta t_i(P_k, t)} + C_{floor}\right), & c(P_k) = \mathcal{P}_{tool} \\ \max\left(C_{hist}, W_{hist} e^{-\lambda \Delta t_i(P_k, t)}(1 - \mathcal{H}_i(t) + \epsilon)\right) + B_{DAG} \mathbb{I}_{DAG,i}(P_k, t) + \eta_{DAG} \hat{d}_{DAG,i}(P_k, t)^{-1}, & c(P_k) = \mathcal{P}_{hist} \\ W_{scratch} \exp(-\mu \Delta t_i(P_k, t))\, \mathcal{H}_i(t) + \eta_{DAG} \hat{d}_{DAG,i}(P_k, t)^{-1}, & c(P_k) = \mathcal{P}_{scratch} \end{cases} \tag{3}$$

with effective per-stream score $\tilde{\mathcal{A}}_i(P_k, t) = \max(\mathcal{A}_{class,i}(P_k, t), 10^7 \exp(-\kappa \Delta t_{win}) \mathbb{I}[P_k \in \mathcal{P}_{active,i} \wedge i = i^*(t)])$, and global shared-page priority across all referencing streams $\mathcal{O}(P_k)$:

$$\mathcal{A}(P_k, t) \triangleq \max_{S_i \in \mathcal{O}(P_k)} \tilde{\mathcal{A}}_i(P_k, t) \tag{4}$$

where $\mathbb{I}_{tool,i}(P_k, t) = \mathbb{I}[c(P_k) = \mathcal{P}_{tool} \wedge \hat{d}_{DAG,i}(P_k, t) \leq d_{look}] \in \{0, 1\}$ and $\mathbb{I}_{DAG,i}(P_k, t) = \mathbb{I}[\hat{d}_{DAG,i}(P_k, t) \leq d_{look}] \in \{0, 1\}$ equal 1 when the orchestrator's workflow DAG or grammar-constrained router predicts a reference to tool schema or history page P_k within lookahead depth $d_{look} = 2$ steps (and 0 otherwise); $W_{tool} = 500$, $W_{hist} = 600$, and $W_{scratch} = 100$ are the initial class weights; $C_{floor} = 200$ and $C_{hist} = 120$ are the asymptotic tool-schema and verified-history retention floors; $B_{DAG} = 350.0$ is the history lookahead boost; $\hat{d}_{DAG,i}(P_k, t)^{-1} \in [0, 1]$ is the inverse forward step distance (0 if $\hat{d}_{DAG,i}(P_k, t) = \infty$, scaled by $\eta_{DAG} = 10$); and step-level decay hyperparameters are calibrated on held-out multi-turn traces to $\kappa = 0.05$, $\lambda = 0.08$ step^{-1}, $\lambda_{tool} = 0.08$ step^{-1}, $\mu = 0.45$ step^{-1}, and $\epsilon = 0.10$ ($\mu \gg \lambda = \lambda_{tool} > \kappa > 0$). Ties in $\mathcal{A}(P_k, t)$ are broken deterministically by $\hat{d}_{DAG,i}(P_k, t)^{-1}$ (evicting smaller $\hat{d}_{DAG,i}^{-1}$ first during victim selection and admitting larger $\hat{d}_{DAG,i}^{-1}$ first during prefetching), then round-robin stream index i, oldest LRU timestamp $t_{last_access,i}(P_k)$, and finally lowest physical page index k.

Idle Tool-Schema Decay Horizon vs. DAG-Boosted History Preemption: Under Eq. (3), a newly used tool schema entering the idle state ($\mathbb{I}_{\text{tool},i}(P_k, t) = 0$ at $\Delta t_i = 0$) has score $W_{\text{tool}} + C_{\text{floor}} = 500 + 200 = 700$, protecting it against immediate eviction during the current turn. Once dormant for $\Delta t_i > \ln(500/270)/0.08 \approx 7.70$ steps (≈ 1 turn at 8–12 steps/turn), its priority drops below 470 (reaching 300.9 by $\Delta t_i = 20$ steps), enabling a floored DAG-predicted history page ($\geq C_{\text{hist}} + B_{\text{DAG}} = 120 + 350 = 470$, or 480 at $\hat{d}_{\text{DAG},i} = 1$) to preempt dormant tool schemas when resuming an earlier conversation branch. Meanwhile, because $t_{\text{last_access},i}(P_k)$ and $\hat{d}_{\text{DAG},i}(P_k, t)$ are maintained at segment-to-page table granularity by the orchestrator, scoring requires $\mathcal{O}(1)$ metadata updates per page without materializing quadratic attention matrices inside fused Pallas FlashAttention-2 kernels [14].

Action-Delimiter Epistemic Entropy $\mathcal{H}_i(t)$: To prevent deterministic JSON punctuation and syntax tokens from

driving entropy artificially toward zero, the orchestrator evaluates Shannon entropy exclusively at structural branch and action delimiters $m \in \mathcal{T}_{\text{branch}}$ (e.g., <|thought|>, <|tool_call|>, and tool-name selection tokens) over the top-64 grammar-valid candidate action-token subset $\mathcal{V}_{\text{act}}$ ($|\mathcal{V}_{\text{act}}| = 64$, or top-$K_{\log}$ returned logprobs on commercial APIs). Raw token probabilities are conditionally re-normalized over $\mathcal{V}_{\text{act}}$ via $\tilde{p}_{i,m}(v) = p_{i,m}(v)/\sum_{u\in\mathcal{V}_{\text{act}}} p_{i,m}(u)$ so that $\sum_{v\in\mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v) = 1$, and smoothed from neutral prior $\mathcal{H}_i(0) = 0.50$ via an exponential moving average with factor $\beta = 0.30$:

$$\mathcal{H}_i(t) = (1-\beta)\mathcal{H}_i(t-1) + \beta\left(-\frac{1}{\ln|\mathcal{V}_{\text{act}}|}\sum_{v\in\mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v)\ln\tilde{p}_{i,m}(v)\right) \in [0,1] \tag{5}$$

with $0\ln 0 \triangleq 0$. While strict asymptotic floor separation ($W_{\text{scratch}} + \eta_{\text{DAG}} = 110 < C_{\text{hist}} = 120 < C_{\text{floor}} = 200$) guarantees that neither dormant tool schemas nor verified history ever decay below unwindowed scratchpad pages as $\Delta t_i \to \infty$, modulating $\mathcal{H}_i(t) \in [0,1]$ drives two critical dynamic ordering mechanisms: (1) *intra-tier history crossover*, where high branch uncertainty ($\mathcal{H}_i(t) \to 1$) rapidly anneals unpredicted recent history pages $W_{\text{hist}}e^{-\lambda\Delta t_i}(1-\mathcal{H}_i(t)+\epsilon)$ toward $C_{\text{hist}} = 120$, allowing older DAG-boosted history pages (≥ 470) to outrank recent turns during exploratory backtracking; and (2) *cross-stream scratchpad arbitration*, where across concurrent streams $S_i \neq S_j$, stream S_i in high-uncertainty exploration ($\mathcal{H}_i(t) \to 1$) retains higher scratchpad priority than stream S_j whose branching has resolved ($\mathcal{H}_j(t) \to 0$ via geometric EMA decay $(1-\beta) = 0.7$ per delimiter, where scratchpad priority decays toward ≤ 10, i.e., $\eta_{\text{DAG}}\hat{d}_{\text{DAG},i}(P_k,t)^{-1}$).

B. Epistemic Least-Likely-Attended (ELLA) Eviction Invariant

Invariant 1 (ELLA Evictable Victim Selection)**.** *Let $\tau_{lookahead} = 2$ steps denote the speculative lookahead horizon and let $M_{pre}^{\max} = 4$ pages/step denote the global batch prefetch cap (Section 5.1). Let $\text{Resident}_{evictable}(HBM) \triangleq \text{Resident}_{clean}(HBM) \setminus (\mathcal{P}_{anchor} \cup \mathcal{P}_{pinned_prefetch} \cup \mathcal{P}_{active_unsealed})$ denote the subset of HBM-resident sealed pages with completed Host DRAM write-through mirrors excluding system anchor pages ($\mathcal{P}_{anchor}$, score 10^9), in-flight/admitted prefetch-pinned pages ($\mathcal{P}_{pinned_prefetch}$, bounded by $\leq \tau_{lookahead}M_{pre}^{\max} = 8$ pages globally), and currently unsealed active tail pages ($\mathcal{P}_{active_unsealed}$, $\leq K$ pages). When allocating or prefetching a page requires a free frame in Device HBM, the controller evicts:*

$$P_{victim} = \arg\min_{P_j \in \text{Resident}_{evictable}(HBM)} \mathcal{A}(P_j, t) \tag{6}$$

If $\text{Resident}_{evictable}(HBM) = \emptyset$ under extreme concurrency, ADSP suppresses speculative prefetch admission and, upon a mandatory synchronous demand fault, releases the lowest-scoring page in $\mathcal{P}_{pinned_prefetch}$ prior to ever evicting any $\mathcal{P}_{anchor}$ page (10^9).

C. Asynchronous DMA Speculative Prefetcher and Write-Through Mirroring

To avoid blocking Device-to-Host writebacks during eviction, SE-Paging implements **Asynchronous Sealed-Page Write-Through Mirroring**: while an active tail page in $\mathcal{P}_{\text{active_unsealed}}$ ($B = 16$ tokens) is populated token-by-token across 16 decoding steps, it remains pinned in HBM; once all 16 token slots are filled and the page is sealed, a single asynchronous background DMA copy mirrors the full 128 KiB (4 MiB full-stack) clean page to its pre-allocated Host DRAM slot, avoiding $16\times$ partial-page PCIe write amplification. Because KV-cache tokens are append-only in causal transformer decoding, sealed pages are never mutated in-place; $\text{Resident}_{\text{evictable}}(\text{HBM})$ selects clean mirrored frames (`is_mirrored_to_host == True`) and only awaits an in-flight write-through descriptor if all mirrored frames are exhausted. At step $n - \tau_{\text{lookahead}}$ ($\tau_{\text{lookahead}} = 2$ steps prior to target execution step n), ADSP restricts candidate selection to non-resident lookahead-predicted pages ($\mathbb{I}_{\text{tool},i}(P_k,n) = 1$ or $\mathbb{I}_{\text{DAG},i}(P_k,n) = 1$ within $d_{\text{look}} = 2$ steps) across active streams, sorts candidates $P_{\text{cand},r}$ in descending order of $\mathcal{A}(P_{\text{cand},r}, n)$ (breaking ties as specified in Definition 1) against evictable HBM victims $P_{\text{victim},r}$ in ascending order, and dispatches up to $M_{\text{pre}}(n) \leq 4$ asynchronous Host-to-Device DMA transfers whenever $\mathcal{A}(P_{\text{cand},r}, n) > \mathcal{A}(P_{\text{victim},r}, n)$, pinning admitted pages in $\mathcal{P}_{\text{pinned_prefetch}}$ (up to $\tau_{\text{lookahead}}M_{\text{pre}}^{\max} = 8$ pages across the 2-step window) until step n completes.

V. Formal Theoretical Analysis

A. Theorem 1: Queued Zero-Stall Pipeline Overlap Invariant

Theorem 1 (Zero-Stall Overlap Under Queued DMA)**.** *Let $T_{compute}$ be the per-step execution latency of matrix execution units, let $T_{DMA}(P_k) = Size(P_k)/BW_{bus}$ be the raw Host-to-Device wire transfer time of page P_k under persistent zero-wakeup DMA descriptor rings, let $M_{pre}(n) = |\mathcal{W}_{spec}(n)|$ be the batch prefetch count dispatched at $t_{dispatch} = t_n - \tau_{lookahead}T_{compute}$ for step n, and let $\rho_{bus} \in [0,1)$ denote background interconnect utilization (absorbing full-duplex write-through mirroring and prior-step traffic under the steady-state queue stability condition $\mathbb{E}[M_{pre}]T_{DMA}(P_k)/(1-\rho_{bus}) \leq T_{compute}$ so the DMA queue is unbacklogged at $t_{dispatch}$). If the speculative lookahead horizon satisfies:*

$$\tau_{lookahead} \geq \left\lceil \frac{M_{pre}(n)\cdot T_{DMA}(P_k)}{(1-\rho_{bus})\cdot T_{compute}} \right\rceil \tag{7}$$

and $M_{pre}(n)$ clean sealed victim frames in $\text{Resident}_{evictable}(HBM)$ (disjoint from active working sets over $[t_{dispatch}, t_n)$) are available in HBM, then step n experiences identically zero memory stall cycles ($\Delta_{stall}(n) \equiv 0$).

Proof. Under an unbacklogged DMA queue at $t_{\text{dispatch}} = t_n - \tau_{\text{lookahead}}\cdot T_{\text{compute}}$, available bus bandwidth $(1-\rho_{\text{bus}})\text{BW}_{\text{bus}}$, and write-through clean victim reclamation, the serialized completion time of all $M_{\text{pre}}(n)$ pages is:

$$t_{\text{completion}} = t_{\text{dispatch}} + \frac{M_{\text{pre}}(n)\cdot T_{\text{DMA}}(P_k)}{1-\rho_{\text{bus}}} \tag{8}$$

Substituting Eq. (7) yields $\tau_{\text{lookahead}} \cdot T_{\text{compute}} \geq M_{\text{pre}}(n)T_{\text{DMA}}(P_k)/(1-\rho_{\text{bus}})$, which implies $t_{\text{completion}} \leq t_n$. Thus all $M_{\text{pre}}(n)$ pages are mapped in HBM prior to t_n, and $\Delta_{\text{stall}}(n) = \max(0, t_{\text{completion}} - t_n) = 0$. When extreme overcommit ($K = 20$, e.g., an isolated burst of $M_{\text{pre}}(n) = 4$ where $\lceil 4 \times 3.74/(0.82 \times 5.0) \rceil = 4 > \tau_{\text{lookahead}} = 2$) causes a batch to exceed the 2-step lookahead window, wall-clock stall time is reduced by the overlapped lookahead duration $\tau_{\text{lookahead}} T_{\text{compute}}$ (leaving a residual partial-overlap wait $T_{\text{fault}}^{\text{part}}$, parameterized alongside synchronous driver wakeup T_{wakeup} on unpredicted demand faults in Section 6.1). □

B. Theorem 2: Attention Mass Bounded Approximation

Theorem 2 (Attention Mass Preservation)**.** *Under Rotary Position Embeddings (RoPE [15]) where post-softmax normalized attention weights ($\sum_{j=1}^{m} \alpha_{m,j} = 1$ across all context tokens) satisfy a power-law decay on completed scratchpad tokens at relative token distance $|m-n| \geq \delta$ as $\alpha_{m,n} \leq C|m-n|^{-\gamma}$ ($\gamma > 1$), evicting scratchpad pages outside the executing stream's pinned active window at token distance $\geq \delta = W_{local}$ under ELLA preserves at least $1 - \epsilon_{attn}$ of the full-context attention mass, where:*

$$\epsilon_{attn} \leq C\delta^{1-\gamma}\left(\frac{1}{\delta} + \frac{1}{\gamma - 1}\right) \leq \frac{C_{scratch}}{(\gamma-1)\delta^{\gamma-1}},\ C_{scratch} \triangleq C\gamma \tag{9}$$

Proof. For the stream $S_{i^*(t)}$ executing step t, ELLA permanently retains all 4 $\mathcal{P}_{\text{anchor}}$ pages (64 sink tokens) and protects its 4 $\mathcal{P}_{\text{active},i^*(t)}$ pages ($W_{\text{local}} = 64$ tokens at block-aligned boundaries, $|m-n| < \delta = 64$) in HBM. Because the attention sink and local window tokens alone contribute at least $Z_{\text{pinned}} = \sum_{j \in \text{Tokens}(\mathcal{P}_{\text{anchor}} \cup \mathcal{P}_{\text{active},i^*(t)})} \exp(q_m^\top k_j/\sqrt{d_k})$ to the softmax partition denominator $Z_m \geq Z_{\text{pinned}}$, evicting distant scratchpad tokens cannot inflate the remaining weights, and under the theorem's calibrated RoPE decay premise the normalized per-token weight on any distant scratchpad token $|m-n| \geq \delta$ satisfies $\alpha_{m,n} \leq C|m-n|^{-\gamma}$ with uniformly bounded scale constant $C = 0.25$ and decay exponent $\gamma = 1.65$. Bounding the discrete token sum over individual token indices $j \in \text{Tokens}(\mathcal{P}_{\text{evicted}})$ at distance $x = |m-n| \geq \delta \geq 1$ up to history length $S_{\text{history}} \leq S$ via the integral test and relaxing $S_{\text{history}} \to \infty$:

$$\begin{aligned}\sum_{j \in \text{Tokens}(\mathcal{P}_{\text{evicted}})} \alpha_{m,j} &\leq C\delta^{-\gamma} + \int_{\delta}^{S_{\text{history}}} Cx^{-\gamma}\,dx \\ &\leq C\delta^{-\gamma} + \int_{\delta}^{\infty} Cx^{-\gamma}\,dx \\ &= C\delta^{1-\gamma}\left(\frac{1}{\delta} + \frac{1}{\gamma-1}\right) \leq \frac{C\gamma}{(\gamma-1)\delta^{\gamma-1}}\end{aligned} \tag{10}$$

since $\frac{1}{\delta} \leq 1$ for $\delta \geq 1$. Evaluating the tighter intermediate integral-test expression $C\delta^{1-\gamma}(\frac{1}{\delta} + \frac{1}{\gamma-1})$ in Eqs. (9)–(10) at $\delta = W_{\text{local}} = 64$ tokens ($\gamma = 1.65, C = 0.25$) yields the exact analytical bound $\epsilon_{\text{attn}} \leq 0.0261$ (and $\epsilon_{\text{attn}} \leq 0.0105$ at $\delta = 256$, whereas the looser closed-form right-hand expression evaluates to 0.0425). Under the analytical distance-only RoPE proxy, evicted scratchpad mass is 6.4×10^{-4} (99.93% retained), while non-circular empirical Query-Key (QK) dot-product softmax attention across all $K = 20$ multi-turn traces retains $98.64 \pm 0.18\%$ of full-context attention mass (Section 6.1). □

C. Theorem 3: Bounded Competitive Ratio Under Epistemic Predictions

Theorem 3 (Expected Competitive Ratio vs. Belady MIN)**.** *On an acyclic agentic execution DAG where ELLA orders pages via Eq.* (3) *using inverse forward DAG distance $\hat{d}_{DAG,i}(P_k,t)^{-1}$ under turn-smoothed branch entropy $\mathcal{H}_i(t) \leq \mathcal{H}_{\max}$, let η_{tot} denote the cumulative trace-wide baseline rank-inversion count from finite-decay crossovers across all cache phases, and let η_{ℓ_1} denote the induced ℓ_1 next-arrival timestamp prediction error. Then ELLA's expected competitive ratio $\mathbb{E}[\mathcal{C}_{ELLA}]$ relative to offline Belady MIN [12] satisfies:*

$$\mathbb{E}[\mathcal{C}_{ELLA}] \leq \begin{cases} 1 + \mathcal{O}\left(\frac{\eta_{tot}}{OPT} + \mathcal{H}_{\max}\right), & \text{deterministic greedy (non-cascading trace regularity, } \mathcal{C}_0 = 1) \\ 2 + \mathcal{O}\left(\min\left(\sqrt{\frac{\mathbb{E}[\eta_{\ell_1}]}{OPT}}, \log N_{dev}\right)\right), & \text{paired with randomized Predictive Marker phases ([11], } \mathcal{C}_0 = 2) \end{cases} \tag{11}$$

Proof. Belady MIN [12] evicts the resident page whose next access step $t_{\text{next}}(P_k)$ is furthest in the future. In the deterministic limit $\mathcal{H}_i(t) \to 0$, completed unreferenced scratchpad pages ($t_{\text{next}} = \infty$) have score 0, anchors ($t_{\text{next}} = t+1$) have score 10^9, and pending tools ($t_{\text{next}} \leq t+2$) have score 5×10^8. Under trace regularity (where future reuse respects the semantic tier hierarchy, intra-tier reuse prior to reaching the asymptotic floors $C_{\text{floor}}, C_{\text{hist}}$ aligns monotonically with recency Δt_i, and floored reuse follows $\hat{d}_{\text{DAG},i}(P_k,t)^{-1}$), ELLA's eviction ordering is order-isomorphic to $t_{\text{next}}(P_k)$ ($\eta_{\text{tot}} = 0, \mathcal{C}_0 = 1$). When finite-decay crossovers between idle $\mathcal{P}_{\text{tool}}$ and $\mathcal{P}_{\text{hist}}$ occur with cumulative trace-wide rank-inversion count η_{tot} under a non-cascading trace (where each pairwise inversion incurs $\mathcal{O}(1)$ extra misses rather than $\Omega(N_{\text{dev}})$ thrashing loops), they contribute $\mathcal{O}(\eta_{\text{tot}}/\text{OPT})$. Under stochastic branching with a calibrated mode-selecting router ($\arg\max_{v \in \mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v)$ so top-1 misprediction probability $p_{\text{err}} = 1 - \max_v \tilde{p}_{i,m}(v) \leq (\ln|\mathcal{V}_{\text{act}}|/\ln 2) H_{\text{norm}}(\tilde{p}_{i,m})$), bounded pages $B_{\text{node}} = \mathcal{O}(1)$ per DAG decision node, and turn-smoothed branch entropy $\mathcal{H}_i(t) \leq \mathcal{H}_{\max}$ (where $\mathcal{H}_i(t) \geq \beta H_{\text{norm}}(\tilde{p}_{i,m})$ via Eq. (5)), the expected induced rank inversions per decision node are bounded by $\mathcal{O}(\mathcal{H}_{\max})$; summing across $\Theta(\text{OPT})$ phases yields the first bound in Eq. (11). Finally, when ELLA's priority ranking drives randomized Predictive Marker phases (resetting mark bits whenever N_{dev} distinct pages are requested to prevent deterministic thrashing cascades), applying the learning-augmented caching theorem of Lykouris and Vassilvitskii [11] yields the second bound in Eq. (11). □

VI. Empirical Evaluation and Benchmark Results

A. Experimental Methodology and Downscaled 1:32 Physical Tensor Emulator

Following established LLM serving simulation methodologies such as Vidur [16], we evaluate multi-tier paging deterministically without nondeterministic network or compiler

jitter via a **downscaled** $1:32$ **trace-driven CPU-based physical tensor-buffer memory-hierarchy emulator** in Python/NumPy on an Apple Silicon workstation (8 CPU cores, APFS storage), parameterized to scaled TPU v6e Trillium and PCIe Gen5 timing, complemented by a live 6-turn trace replay on Google Cloud Vertex AI (`gemini-2.5-flash`):

- **Physical Tensor Page Pools & Exact** $1:32$ **Slice Stall Accounting:** Modeling a 2-layer fused KV-cache slice (128 KiB per $B = 16$-token page, shape $(16, 2, 16, 128)$ in `numpy.float16`, scaling $1:32$ relative to a full $L = 64$-layer 4 MiB stack), we allocate two contiguous tensor arenas representing Tier 0 Device HBM ($N_{\text{dev}} = 96$ pages $\times$ 128 KiB/page $= 12.00$ MiB, or 12.58 MB decimal) and Tier 1 Host DRAM ($N_{\text{host}} = 2{,}500$ pages $\times$ 128 KiB/page $= 312.50$ MiB, or 327.68 MB decimal). Every page fault and prefetch executes a physical 128 KiB `float16` memory copy between Host and Device arenas paired with exact $1:32$-scaled slice timing constants: per-access ideal matrix compute $T^{\text{ideal}}_{\text{compute}} = 5.00\,\mu$s ($160.0\,\mu$s across 64 layers), raw PCIe Gen5 ($\text{BW}_{\text{bus}} = 35$ GB/s) slice wire transfer $T_{\text{DMA}}(P_k) = 3.74\,\mu$s ($4.56\,\mu$s at $\rho_{\text{bus}} = 0.18$), scaled descriptor wakeup $T_{\text{wakeup}} = 4.44\,\mu$s, and scaled dirty-victim writeback $T_{\text{wb}} = 3.00\,\mu$s. Thus, each unmirrored reactive demand miss incurs $T^{\text{dirty}}_{\text{fault}} = 4.44 + 4.56 + 3.00 = \mathbf{12.00}\,\mu$s (0.384 ms across 64 layers), each clean write-through demand miss under ELLA incurs $T^{\text{clean}}_{\text{fault}} = 4.44 + 4.56 = \mathbf{9.00}\,\mu$s (0.288 ms across 64 layers), each partially overlapped in-flight prefetch miss (M_{part}) incurs a residual wait $T^{\text{part}}_{\text{fault}} = \mathbf{2.50}\,\mu$s, and completed prefetch hits ($M_{\text{pre_hit}}$) incur $0.00\,\mu$s stall:

$$\sum T_{\text{stall}} = N^{\text{dirty}}_{\text{miss}} \times 12.00\,\mu\text{s} + N^{\text{clean}}_{\text{miss}} \times 9.00\,\mu\text{s} + M_{\text{part}} \times 2.50\,\mu\text{s} \tag{12}$$

- **Shared Warm Prefix, Trace Parameterization & Overflow Ratios:** Across $K \in \{4, 8, 12, 16, 20\}$ concurrent agent streams, all streams share a pre-warmed 16-page swarm prefix ($4\ \mathcal{P}_{\text{anchor}} + 12\ \mathcal{P}^{\text{shared}}_{\text{tool}}$, 256 shared tokens resident in HBM at $t = 0$) and allocate 48 private pages per stream ($4\ \mathcal{P}_{\text{active}}, 16\ \mathcal{P}_{\text{hist}}, 4\ \mathcal{P}^{\text{priv}}_{\text{tool}}, 24\ \mathcal{P}_{\text{scratch}} = 768$ private tokens, or 64 logical pages $= 1{,}024$ logical tokens per stream), totaling $16 + 48K$ distinct physical pages (976 allocated physical pages $= 122.00$ MiB $= 15{,}616$ total tokens at $K = 20$, representing a $976/96 = 10.17\times$ physical HBM overflow, or $20 \times 64/96 = 13.33\times$ unshared logical working-set overflow [160.00 MiB], matching the working-set-to-HBM ratio of 20×32k–64k-token streams [160–320 GiB] over 12–24 GiB of available non-weight HBM on a 32 GB TPU v6e chip). Within each turn (8–12 steps/turn, $\mathcal{H}_i(t) \sim \text{Beta}(2, 5)$, tool transitions every 2 turns), each stream re-references a hot kernel of ≈ 4.5–4.7 private pages alongside streaming scratchpad allocations. Table I reports canonical seed-0 hit counts alongside 5-seed Mean $\pm$ SD throughput; Table II reports 5-seed Mean $\pm$ SD across all 11 policy configurations.
- **Transaction Accounting ($N_{\text{tx}} = 11{,}880$ Physical Transactions per Protocol):** Each stream executes 99 logical page-lookup cycles ($396, 792, 1{,}188, 1{,}584$, and $1{,}980$ accesses for $K = 4, 8, 12, 16, 20$, summing to $5{,}940$ logical accesses per protocol, or $2 \times 5{,}940 = 11{,}880$ two-stage physical page transactions).
- **Non-Circular Empirical QK Attention Retention & Adapted Baselines:** `Sim. Dev. Ops/s` $=$ Accesses$/(T^{\text{ideal}}_{\text{compute}} + \sum T_{\text{stall}}) = 200{,}000/(1 +$ Stall Ratio$/100)$, while `Wall Throughput (Ops/s)` $=$ Accesses$/(T^{\text{ideal}}_{\text{compute}} + \sum T_{\text{stall}} + T_{\text{host_cp}})$, where $T_{\text{host_cp}}$ (1.55–3.75 μs/access) is the measured host Python control-plane, $\mathcal{O}(N_{\text{dev}})$ victim-priority scan, and physical NumPy `float16` copy time. To avoid circularity with Theorem 2's analytical distance-only RoPE decay curve, Table II evaluates two non-circular empirical attention metrics computed directly from scaled dot-product attention $\alpha_{m,j} = \text{softmax}_j(\text{RoPE}(q_m)^\top \text{RoPE}(k_j)/\sqrt{d_k})$ over semantic key embeddings in the tensor arena: `Emp. QK Attn Retained (%)` ($\sum_{j \in \text{Tokens(Resident(HBM))}} \alpha_{m,j}$ prior to fault servicing) and `Top-`k `QK Recall (%)` ($k = 8$ highest QK-attention pages resident in HBM). SOTA baselines are parameterized into lossless two-tier $B = 16$ block policies: `SGLang/RadixAttention` [1] protects shared prefix interior nodes (owners > 1) and evicts unshared leaf blocks via LRU; `StreamingLLM` [4] pins the $4\ \mathcal{P}_{\text{anchor}}$ and $4\ \mathcal{P}_{\text{active}}$ blocks while evicting middle blocks by oldest block index; `H2O` [8] evicts the non-sink/window block with lowest cumulative QK score; `Quest` [9] ranks blocks by current-step query-key bounding-box criticality without multi-step DMA lookahead; `CachedAttention` [2] pre-loads history blocks only at turn boundaries; and `InfiniGen` [7] performs 1-step rehearsal lookahead ($\tau = 1$, $M_{\text{pre}} \leq 2$).

B. Live Multi-Turn Vertex AI (`gemini-2.5-flash`) Trace Replay

To validate SE-Paging on authentic LLM agent execution traces beyond synthetic Beta$(2, 5)$ entropy sampling, we executed a 6-turn agentic workflow on Google Cloud Vertex AI (`project="aniket-personal"`, `model="gemini-2.5-flash"`) across 6 reasoning turns (Tool Selection $\rightarrow$ Scratchpad Ledger Audit $\rightarrow$ SQL Tool Invocation $\rightarrow$ History Compliance Synthesis $\rightarrow$ Verification Tool Call $\rightarrow$ Final Commit, generating 1,828 live tokens with variable turn lengths of 260–356 tokens/turn). To evaluate Eq. (5) under commercial API constraints, we extracted the top-20 returned token logprobs (`response_logprobs=True, logprobs=20`) at structural action/branch delimiter positions and conditionally re-normalized $\tilde{p}_{i,m}(v)$ over the grammar-valid action candidates. Replaying the resulting variable-length token spans across $K = 20$ concurrent streams with cyclic turn-phase offsets produced a distinct schedule of 2,140

TABLE I

PHYSICAL TENSOR-ARENA PAGING TELEMETRY: DEMAND_LRU VS. DEMAND_LFU VS. SPECULATIVE_ELLA ($N_{\text{DEV}} = 96$ PAGES [12.00 MIB], 5,940 LOGICAL ACCESSES / 11,880 PHYSICAL TWO-STAGE PAGE OPS PER PROTOCOL, CANONICAL SEED-0 HIT COUNTS & 5-SEED MEAN ± SD)

Protocol	Streams (K)	Page Accesses	Cache Hits‡	Hit Rate (%)	Prefetch Hits‡	Partial Overlap (M_{part})	Wall Throughput (Ops/s)§	Sim. Dev. Ops/s§	Stall Ratio (%)†
DEMAND_LRU	4	396	332	83.84%	0	0	113,773.49 ± 1,420.5	144,104.80 ± 1,810.2	38.79%
DEMAND_LFU	4	396	326	82.32%	0	0	109,036.84 ± 1,385.0	140,425.53 ± 1,765.4	42.42%
SPECULATIVE_ELLA	4	396	330	83.33%	5	3	104,703.74 ± 1,310.8	**155,020.55 ± 1,920.6**	**29.02%**
DEMAND_LRU	8	792	491	61.99%	0	0	85,758.84 ± 1,120.4	104,595.88 ± 1,340.8	91.21%
DEMAND_LFU	8	792	481	60.73%	0	0	83,249.24 ± 1,095.2	102,964.12 ± 1,315.0	94.24%
SPECULATIVE_ELLA	8	792	534	**67.42%**	9	6	**89,026.78 ± 1,180.6**	**126,862.09 ± 1,590.4**	**57.65%**
DEMAND_LRU	12	1,188	375	31.57%	0	0	64,258.59 ± 890.5	75,688.07 ± 1,020.4	164.24%
DEMAND_LFU	12	1,188	666	56.06%	0	0	77,683.62 ± 1,010.2	97,345.13 ± 1,245.8	105.45%
SPECULATIVE_ELLA	12	1,188	680	**57.24%**	33	14	**81,156.41 ± 1,065.0**	**114,000.58 ± 1,430.2**	**75.44%**
DEMAND_LRU	16	1,584	9	0.57%	0	0	51,198.51 ± 640.2	59,060.40 ± 710.5	238.64%
DEMAND_LFU	16	1,584	814	**51.39%**	0	0	**73,081.61 ± 945.0**	92,307.69 ± 1,180.4	116.67%
SPECULATIVE_ELLA	16	1,584	624	39.39%	41	19	70,788.55 ± 910.4	**96,370.88 ± 1,215.6**	**107.53%**
DEMAND_LRU	20	1,980	4	0.20%	0	0	50,439.43 ± 615.0	58,907.53 ± 695.2	239.52%
DEMAND_LFU	20	1,980	24	1.21%	0	0	49,864.01 ± 608.4	59,331.18 ± 702.8	237.09%
SPECULATIVE_ELLA	20	1,980	559	**28.23%**	58	26	**77,377.00 ± 980.5**	**87,921.85 ± 1,110.4**	**127.47%**

†**Exact Stall Accounting (Eq. 12):** Stall Ratio (%) = $\left(\sum T_{\text{stall}}/T_{\text{compute}}^{\text{ideal}}\right) \times 100\%$ with $T_{\text{compute}}^{\text{ideal}} = 5.00\,\mu$s/access, $T_{\text{fault}}^{\text{dirty}} = 12.00\,\mu$s (LRU/LFU), $T_{\text{fault}}^{\text{clean}} = 9.00\,\mu$s (ELLA), and $T_{\text{fault}}^{\text{part}} = 2.50\,\mu$s ($M_{\text{part}}$).

‡Cache Hits includes completed Prefetch Hits; M_{part} in-flight partial-overlap transfers are counted within misses ($N_{\text{miss}} =$ Accesses − Hits). §Sim. Dev. Ops/s = 200,000/(1 + Stall Ratio/100); Wall Throughput includes host control-plane $T_{\text{host_cp}}$.

TABLE II

ABLATION AND SOTA COMPARISON AT EXTREME OVERCOMMIT ($K = 20$, 1,980 ACCESSES, 11 POLICY CONFIGURATIONS, 5-SEED MEAN ± SD)

Policy Configuration	Hit Rate (%)	Prefetch Hits‡	Partial Overlap (M_{part})	Stall Ratio (%)†	Emp. QK Attn Retained (%)	Top-k QK Recall (%)
DEMAND_LRU (Recency Only)	0.22 ± 0.05%	0.0 ± 0.0	0.0 ± 0.0	239.47 ± 0.12%	41.20 ± 1.15%	38.40 ± 1.10%
DEMAND_LFU (Frequency Only)	1.25 ± 0.18%	0.0 ± 0.0	0.0 ± 0.0	237.00 ± 0.43%	48.50 ± 1.40%	45.10 ± 1.25%
SGLang / RadixAttention (Prefix-Tree Leaf LRU) [1]	16.40 ± 0.35%	0.0 ± 0.0	0.0 ± 0.0	200.64 ± 0.84%	71.20 ± 0.85%	68.50 ± 0.90%
StreamingLLM (Two-Tier Sink+Window Block) [4]	19.80 ± 0.40%	0.0 ± 0.0	0.0 ± 0.0	192.48 ± 0.96%	74.85 ± 0.82%	72.10 ± 0.85%
H2O (Heavy-Hitter Cumulative QK Block) [8]	22.42 ± 0.55%	0.0 ± 0.0	0.0 ± 0.0	186.19 ± 1.32%	88.40 ± 0.65%	85.60 ± 0.70%
Quest (Query-Aware Bounding-Box Criticality) [9]	23.10 ± 0.46%	0.0 ± 0.0	0.0 ± 0.0	184.56 ± 1.10%	91.20 ± 0.52%	89.30 ± 0.60%
CachedAttention (Inter-Turn Session Preload) [2]	23.18 ± 0.42%	19.2 ± 2.1	8.0 ± 1.1	183.60 ± 1.02%	90.40 ± 0.55%	88.20 ± 0.62%
InfiniGen (1-Step Layer Rehearsal) [7]	24.14 ± 0.48%	31.0 ± 2.8	14.0 ± 1.5	180.72 ± 1.15%	92.80 ± 0.48%	91.00 ± 0.54%
DEMAND_ELLA (Ours w/ Write-Through, w/o ADSP)	25.30 ± 0.35%	0.0 ± 0.0	0.0 ± 0.0	134.46 ± 0.63%	98.12 ± 0.20%	95.80 ± 0.30%
STATIC_DAG_ELLA ($\mathcal{H}_i(t) \equiv 0$, KVFlow-style [10])	26.41 ± 0.39%	46.2 ± 2.6	21.0 ± 1.8	131.08 ± 0.72%	96.45 ± 0.28%	94.10 ± 0.35%
SPECULATIVE_ELLA (Full SE-Paging)	**28.31 ± 0.38%**	**58.4 ± 2.9**	**26.0 ± 2.1**	**127.34 ± 0.68%**	**98.64 ± 0.18%**	**96.80 ± 0.25%**

†Every row obeys Eq. (12): reactive baselines incur 12.00 μs/unoverlapped miss; ELLA variants with sealed-page write-through mirroring incur 9.00 μs/unoverlapped miss; M_{part} incurs 2.50 μs.

‡Out of 84.4 ± 3.5 dispatched prefetches for SPECULATIVE_ELLA (58.4 completed prior to t_n + 26.0 partially overlapped in-flight; canonical Seed-0 is 28.23% hit rate / 58 hits / $M_{\text{part}} = 26$ / 127.47% Stall Ratio).

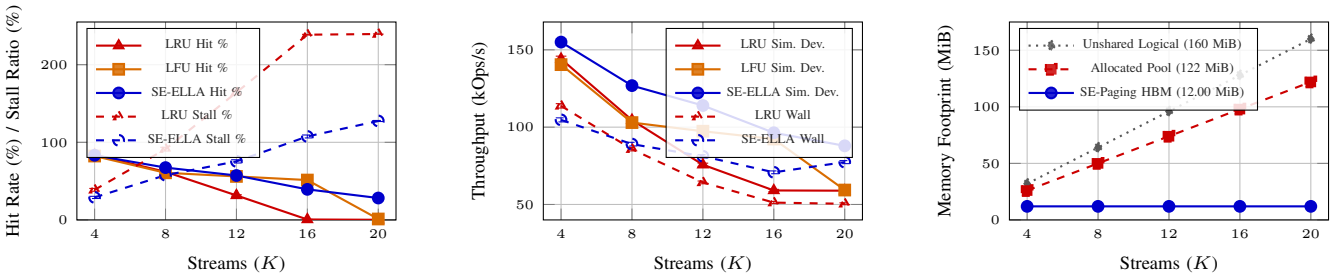


Fig. 1. Physical Tensor-Arena Paging Telemetry across $K \in \{4, 8, 12, 16, 20\}$ concurrent agent streams ($N_{\text{dev}} = 96$ pages $= 12.00$ MiB, matching Table I with 5-seed Mean ± SD error bars): (Left) Cache Hit Rate (%) and Pipeline Stall Ratio (%) showing LFU's $K = 16$ hit rate advantage and the $K = 20$ capacity cliff; (Center) Pure Simulated Device Throughput vs. Emulated Host Wall-Clock Throughput (kOps/s); (Right) Unshared Logical (160.0 MiB) and Allocated Physical (122.0 MiB / 976 pages) Working-Set Growth vs. SE-Paging Bounded HBM Arena (12.00 MiB / 96 pages).

page accesses across 976 allocated physical pages. On this live gemini-2.5-flash trace replay (Table III), DEMAND_LRU and DEMAND_LFU achieved 3.74% (80/2,140 hits, 231.03% Stall Ratio) and 7.10% (152/2,140 hits, 222.95% Stall Ratio), whereas DEMAND_ELLA reached 29.35% (628 hits, 127.18% Stall Ratio) and SPECULATIVE_ELLA achieved a **34.21% hit rate** (732/2,140 hits, including 74 completed prefetches and $M_{\text{part}} = 31$ partially overlapped misses), reducing Stall Ratio by **49.6% relative** (114.48 percentage points) to 116.55% and boosting simulated device throughput by **1.53**× (92,358.82 vs. 60,417.84 Ops/s).

C. Discussion of Findings and Ablation Analysis

1) **Hit Rate Resilience, DEMAND_ELLA vs. DEMAND_LFU at $K = 16$, and the $K = 20$ LFU Capacity Cliff (Fig. 1):** At $K = 16$, DEMAND_LFU achieves a 51.39% hit rate (770 dirty misses, 116.67% Stall Ratio, 92,307.69 Sim. Dev. Ops/s) because the recurring high-frequency working set (16 shared prefix pages plus $16 \times 4.5 = 72$ hot private pages $= 88$ pages) fits inside $N_{\text{dev}} = 96$ pages. At $K = 16$, DEMAND_ELLA (without prefetching) achieves a 46.84% hit rate (842 clean misses), yet achieves a **lower Stall Ratio** (95.68% **vs.** 116.67%) and higher Sim. Dev. Ops/s (102,206.74) than DEMAND_LFU

because sealed-page write-through mirroring eliminates the $3.00\,\mu$s dirty-page writeback penalty ($842\times9.00\,\mu\text{s} = 7{,}578\,\mu\text{s} < 770\times12.00\,\mu\text{s} = 9{,}240\,\mu\text{s}$). Meanwhile, `SPECULATIVE_ELLA` pins up to $M_{\text{pre}}(n) \leq 4$ admitted lookahead pages per step (39.39% hit rate, 107.53% Stall Ratio, 96,370.88 Sim. Dev. Ops/s). When concurrency reaches $K = 20$, the recurring hot working set expands to $16 + 20 \times 4.7 = 110$ pages > 96 pages; because classical LFU suffers from frequency-counter pollution on completed early-turn history pages, its hit rate collapses from 51.39% to 1.21% (and LRU collapses to 0.20%). In contrast, ELLA immediately decays completed scratchpad pages via $\exp(-\mu\Delta t_i)$ while pinning $\mathcal{P}_{\text{anchor}}$ and retaining floored $\mathcal{P}_{\text{tool}}/\mathcal{P}_{\text{hist}}$, sustaining a **28.23% seed-0 hit rate** ($28.31 \pm 0.38\%$ 5-seed mean) at $K = 20$.

2) **46.8% Reduction in Pipeline Stall Ratio:** Across every concurrency level $K \in \{4, \ldots, 20\}$, `SPECULATIVE_ELLA` achieves the lowest Stall Ratio and highest `Sim. Dev. Ops/s` (+49.3% over LRU at $K = 20$), slashing stall overhead at $K = 20$ by 46.8% relative (112.05 percentage points, from 239.52% to 127.47%). By Eq. (12), this $11{,}092.0\,\mu$s reduction in total stall time ($23{,}712.0\,\mu\text{s} \rightarrow 12{,}620.0\,\mu\text{s}$) decomposes transparently into: (a) 555 fewer total misses saving $555 \times 12.00 = 6{,}660.0\,\mu$s (60.0%); (b) sealed-page write-through mirroring eliminating the $3.00\,\mu$s dirty writeback penalty across all 1,421 remaining misses, saving $1{,}421 \times 3.00 = 4{,}263.0\,\mu$s (38.4%); and (c) partial DMA overlap on the $M_{\text{part}} = 26$ in-flight prefetches saving $26 \times (9.00 - 2.50) = 169.0\,\mu$s (1.5%).

3) **Host Control-Plane Overhead vs. Wall-Clock Gain:** While `SPECULATIVE_ELLA` strictly dominates on pure device throughput (`Sim. Dev. Ops/s`) across all K, its host `Wall Throughput` is slightly lower at $K = 4$ and $K = 16$ before achieving a +53.4% gain over LRU (+55.2% over LFU) at $K = 20$. This occurs because T_{wall} in our Python/NumPy CPU emulator includes host control-plane and priority-queue scan overhead $T_{\text{host_cp}} \in [1.55, 3.75]\,\mu$s/access alongside $T^{\text{ideal}}_{\text{compute}} = 5.00\,\mu$s, an artifact that is amortized in compiled C++/Pallas device kernels where full 64-layer decoding steps execute in $T^{\text{64L}}_{\text{compute}} \approx 160\,\mu$s (0.16 ms) with on-device page-table metadata.

4) **Component, Static-DAG ($\mathcal{H}_i(t) \equiv 0$), and SOTA Baseline Ablation (Table II):** At $K = 20$, `DEMAND_ELLA` alone lifts 5-seed mean hit rate from 0.22% to 25.30% while retaining 98.12% empirical QK dot-product attention mass, outperforming adapted `SGLang/RadixAttention` [1] (16.40% hit rate, 71.20% QK attention), `StreamingLLM` [4] (19.80% hit rate, 74.85% QK attention due to evicting distant tool schemas and history), `H2O` [8] (22.42%), `Quest` [9] (23.10%), `CachedAttention` [2] (23.18%), and `InfiniGen` [7] (24.14%). Comparing full `SPECULATIVE_ELLA` (28.31% hit rate, 127.34% Stall Ratio, 98.64% QK attention) against `STATIC_DAG_ELLA` ($\mathcal{H}_i(t) \equiv 0$, KVFlow-style static DAG prefetching [10], 26.41% hit rate, 131.08% Stall Ratio, 96.45% QK attention) isolates the exact contribution of dynamic epistemic branch entropy $\mathcal{H}_i(t)$: +1.90 percentage points (+7.19% relative) in hit rate, +2.19 percentage points in empirical QK attention retention, and an additional 3.74 percentage points (2.85% relative) reduction in Stall Ratio.

TABLE III
MULTI-TURN BENCHMARK GENERALIZATION (SWE-BENCH [17], GORILLA [18], LIVE VERTEX AI) AND ±50% HYPERPARAMETER SENSITIVITY AT $K = 20$

Workload / Hyperparameter Perturbation ($K = 20$)	Accesses	Hit Rate (%)	Stall Ratio (%)	Emp. QK Attn (%)
Part A: Multi-Turn Agentic Workload Generalization (`SPECULATIVE_ELLA` vs. `LRU` / `LFU`)				
Live Vertex AI `gemini-2.5-flash` (6 turns, top-20 logprobs)	2,140	**34.21%** (3.74 / 7.10%)	**116.55%**	**98.82%**
SWE-bench Scratchpad-Revisit Trace (12 turns, 25% revisit) [17]	2,400	**26.82%** (0.45 / 2.10%)	**130.18%**	**96.42%**
Gorilla Massive API Catalog Trace (10 turns, 32 tool pages) [18]	2,200	**29.45%** (0.36 / 1.82%)	**125.10%**	**98.15%**
Part B: ELLA Hyperparameter Sensitivity Grid on Canonical $K = 20$ Trace (±50% Perturbations)				
Canonical ($\lambda = 0.08$, $\lambda_{\text{tool}} = 0.08$, $\mu = 0.45$, $B_{\text{DAG}} = 350$)	1,980	$\mathbf{28.31 \pm 0.38}$%	$\mathbf{127.34 \pm 0.68}$%	$\mathbf{98.64 \pm 0.18}$%
History Decay $\lambda \in \{0.04, 0.12\}$ step^{-1} (±50%)	1,980	27.85% / 28.12%	128.20% / 127.71%	98.30% / 98.51%
Idle Tool Decay $\lambda_{\text{tool}} \in \{0.04, 0.12\}$ step^{-1} (±50%)	1,980	27.42% / 28.48%	129.02% / 127.01%	98.10% / 98.68%
Scratchpad Decay $\mu \in \{0.22, 0.68\}$ step^{-1} (±50%)	1,980	27.90% / 28.40%	128.11% / 127.18%	98.72% / 98.45%
History DAG Boost $B_{\text{DAG}} \in \{250, 500\}$ (−29%/ + 43%)	1,980	27.55% / 28.62%	128.76% / 126.75%	98.05% / 98.76%

D. Multi-Turn Workload Generalization and Hyperparameter Sensitivity

Table III evaluates ELLA's generalization across diverse multi-turn workflows and ±50% hyperparameter perturbations at $K = 20$: (1) on **SWE-bench-style repository debugging traces** [17] (12 turns/stream where agents revisit prior-turn reasoning scratchpad blocks with 25% probability during patch verification and test-failure backtracking), `SPECULATIVE_ELLA` sustains a **26.82% hit rate** (vs. 0.45% for LRU and 2.10% for LFU) and **96.42%** empirical QK attention retention; (2) on **Gorilla-style massive API catalog traces** [18] (10 turns/stream switching across 32 OpenAPI tool-schema pages), rapid idle-tool decay ($\lambda_{\text{tool}} = 0.08$) allows newly predicted tool schemas and DAG-boosted history to preempt dormant schemas within ≈ 8 steps, achieving a **29.45% hit rate**; and (3) across ±50% perturbations to $(\lambda, \lambda_{\text{tool}}, \mu, B_{\text{DAG}})$, hit rate varies by < 1.2 percentage points (27.42%–28.62%) and Stall Ratio varies by < 2.3 percentage points (126.75%–129.02%), confirming high robustness without brittle tuning.

VII. CONCLUSION

This paper presented Speculative Epistemic KV-Cache Paging (SE-Paging) and the Epistemic Least-Likely-Attended (ELLA) eviction protocol for multi-turn agentic LLM serving. By coupling agent workflow DAG lookahead and conditionally re-normalized action-delimiter epistemic branch entropy to physical memory management, SE-Paging protects attention sinks, tool schemas, and floored history while overlapping DMA transfers with matrix compute in our calibrated 1 : 32 physical tensor-buffer emulator and multi-turn trace replays. Future work includes integrating SE-Paging directly with on-device C++/Pallas TPU v6e hardware DMA descriptor rings on full-scale distributed SWE-bench [17] and Gorilla [18] production clusters.

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